# Missing Mass and Minimum Wages: Distributional Effects of Three Minimum Wage Increases in Peru

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*Preliminary draft — do not circulate*

**Abstract**

I study the distributional effects of three minimum wage (MW) increases in Peru—from S/750 to S/1,025 between 2016 and 2022—using a pre-post distributional estimator adapted from Cengiz et al. (2019) for a national MW setting without control jurisdictions. Using ENAHO household survey data covering 9,000–11,700 formal dependent workers per year, I document clear missing mass below the old MW floor and excess mass above the new floor in all three events. Bunching ratios range from 0.70 to 0.83: for every 100 jobs that disappear from below the new minimum, 70–83 reappear above it. Placebo tests at artificial thresholds produce ratios 7 times smaller, confirming the signal is MW-specific. Results are replicated on EPE Lima quarterly data with tighter 6-month event windows. Employment effects cannot be identified: event-study

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pre-trends fail, the Kaitz instrumental variable is weak ($F < 3$), and panel attrition exceeds 76%. Hours show no adjustment ($\leq \pm 1.1$h/week). Wage compression of 3–7 percentage points is present but 41–92% is mechanical composition. The dominant effect of Peru's MW increases within the studied Kaitz range (0.52–0.57) is redistribution of the formal wage distribution, not measurable job destruction.

**JEL Codes**: J31, J38, O17

**Keywords**: minimum wage, bunching estimator, wage distribution, Peru, informality

# Contents


**1 Introduction** **5**

**2 Institutional Context** **6**

2.1 Peru's Minimum Wage System . . . . . 6

2.2 The Three MW Events . . . . . 7

2.3 Department-Level Kaitz Variation . . . . . 7

**3 Data** **8**

3.1 ENAHO Module 500 . . . . . 8

3.2 Summary Statistics . . . . . 8

**4 Empirical Strategy** **9**

4.1 Pre-Post Distributional Estimator . . . . . 9

4.2 Bootstrap Inference . . . . . 11

4.3 Hours DiD . . . . . 11

4.4 Failed Identification Strategies (Appendices B–C) . . . . . 11

**5 Bunching Results** **11**

5.1 Main Estimates . . . . . 11

5.2 Placebo Test . . . . . 13

5.3 Wage Compression . . . . . 14

**6 Heterogeneity** **14**

**7 Hours** **17**

**8 Conclusion** **18**

**A ENAHO Panel 978 — Event C** **20**

**B Event Study — Pre-Trend Violations** **20**

**C IV / OWE — Not Identified** **21**

**D EPEN CIU Annual 2023 — Lee-Saez Bunching** **21**

**E EPE Lima Quarterly Bunching** **21**

# 1 Introduction

Peru raised its minimum wage (*Remuneración Mínima Vital*, RMV) three times between 2016 and 2022, from S/750 to S/1,025—a cumulative 37% nominal increase. Throughout this period, formal employment remained stable and the formality rate recovered from its COVID-19 dip. What happened to the wage distribution? Did workers near the minimum wage floor receive raises, or did they lose their jobs?

This paper uses a pre-post distributional estimator inspired by the framework of Cengiz et al. (2019), adapted for Peru's national MW setting, to answer these questions. The estimator compares the formal wage distribution before and after each MW increase, measuring "missing mass" below the old threshold and "excess mass" at the new one. The ratio of excess to missing mass—the bunching ratio—indicates what fraction of displaced jobs reappear above the new floor.

The key challenge is that Peru has a single national MW with no sub-national variation in the policy. Cengiz et al. (2019) construct their counterfactual by comparing U.S. states with MW changes to states without—a difference-in-differences design. This is not feasible in Peru. Instead, I follow the methodological precedent of Harasztosi and Lindner (2019), who analyze Hungary's national MW using the pre-reform wage distribution as the counterfactual, with a background correction from the upper tail of the distribution.

I analyze three MW events: Event A (S/750→S/850, May 2016), Event B (S/850→S/930, April 2018), and Event C (S/930→S/1,025, May 2022). Using ENAHO household survey data from 2015–2023, covering approximately 10,000 formal dependent workers per year, I find bunching ratios of 0.696, 0.829, and 0.830 for the three events respectively. Bootstrap 95% confidence intervals are [0.567, 0.896], [0.716, 1.016], and [0.716, 0.960]. Events A and C reject $R = 1$ (full redistribution with zero job loss); Event B does not—the data are consistent with both complete redistribution and displacement of up to 28% of affected workers.

Placebo tests at artificial thresholds (S/1,100→S/1,200 and S/1,400→S/1,500) produce

ratios of 0.114 and 0.013, with missing mass 7 times smaller than at the true MW—confirming the bunching signal is MW-specific. Results are stable across bin widths (S/25, S/50, S/100) and across populations. An independent dataset—EPE Lima quarterly data with 6-month event windows—produces consistent ratios (0.73–1.03 vs. ENAHO's 0.70–0.83).

I systematically test three identification strategies for employment effects: department-level event-study DiD, Kaitz instrumental variables at department and province levels, and within-year province IV. All fail. Event-study pre-trends are violated for Events B ($p = 0.007$) and C ($p = 0.017$); Event A has no testable pre-period. The Kaitz IV produces $F$-statistics below 3 in all specifications, far below the Stock-Yogo threshold. The ENAHO panel has 76% attrition with differential selection by treatment group. These failures are not data-quality issues but a structural feature of single-national-MW settings: Kaitz variation across departments reflects poverty gradients with independent secular and seasonal dynamics.

The paper contributes to the MW literature in three ways. First, it provides the first distributional analysis of Peru's MW using three identifiable events, contributing to the growing evidence base from developing countries (see Harasztosi and Lindner, 2019; Saltiel and Urzúa, 2018; Gindling and Terrell, 2021). Second, it documents the systematic failure of regression-based identification in a national-MW setting, offering a methodological cautionary note for researchers working in similar institutional contexts. Third, by decomposing apparent wage compression into mechanical (composition) and genuine components, it clarifies that 41–92% of measured compression reflects MW-induced worker sorting, not spillover wage effects.

# 2 Institutional Context

## 2.1 Peru's Minimum Wage System

The *Remuneración Mínima Vital* (RMV) is set by executive decree and applies to all registered formal dependent workers (*empleados* and *obreros*). It is not indexed to inflation or

productivity and changes only when the government issues a new decree. Enforcement is conducted by SUNAFIL (*Superintendencia Nacional de Fiscalización Laboral*), with inspections concentrated in Lima and large firms.

Peru's labor market is characterized by high informality: approximately 72–75% of employed workers are informal by INEI's comprehensive definition. The MW directly affects roughly 10–15% of the workforce classified as formal dependent workers under INEI's formality variable (`ocupinf == 2`), which requires social security registration or a written employment contract.

## 2.2 The Three MW Events

Table 1 summarizes the three MW changes analyzed in this paper. Each event is separated by at least two years, allowing clean pre-post comparisons using annual ENAHO surveys.

Table 1: Minimum Wage Events Analyzed

| Event | Pre-MW | Post-MW | Change | Effective | Analysis years |
|---|---|---|---|---|---|
| A | S/750 | S/850 | +13.3% | May 2016 | Pre: 2015, Post: 2017 |
| B | S/850 | S/930 | +9.4% | April 2018 | Pre: 2017, Post: 2019 |
| C | S/930 | S/1,025 | +10.2% | May 2022 | Pre: 2021, Post: 2023 |

*Note*: No 2020 survey due to COVID-19. Event C's pre-period (2021) reflects post-pandemic labor market conditions.

## 2.3 Department-Level Kaitz Variation

The Kaitz index (MW divided by the weighted-median formal wage by department) ranges from 0.39 to 1.09 across departments and years. Ica (department 11) has the highest Kaitz (0.94 in 2015), driven by formal agricultural workers on agro-export farms earning near-minimum monthly wages despite full-time hours (median 48 hours/week). Huancavelica

(department 09) has a low Kaitz ($\approx 0.50$) because approximately 85% of its formal workers are public-sector employees earning above the MW.

This variation primarily reflects differences in economic structure—the shares of agriculture and public-sector employment—rather than differential MW exposure. It does not generate sufficient first-stage power for IV identification ($F < 3$ in all events; see Appendix C), and employment pre-trends fail for Events B and C (Appendix B). The Kaitz index is used in this paper as a descriptive measure of MW bindingness, not as an instrument.

# 3 Data

## 3.1 ENAHO Module 500

The primary data source is the *Encuesta Nacional de Hogares* (ENAHO) Module 500 (*Empleo e Ingresos*), conducted annually by INEI. I use cross-sectional data from eight years: 2015–2019, 2021–2023 (no 2020 survey due to COVID-19).

The analysis sample consists of formal dependent workers with monthly wages between S/1 and S/6,000 and positive survey weights. The filter chain is: employed (`ocu500 == 1`), dependent worker (`p507` $\in \{3, 4, 6\}$: *empleado*, *obrero*, *trabajador del hogar*), formal (`ocupinf == 2`), wage $\in (0, 6000)$, weight $> 0$. The wage variable is `p524a1` (monthly cash income from primary occupation). The weight variable is `fac500a`. Department is identified from the first two digits of `ubigeo`.

## 3.2 Summary Statistics

Table 2 presents summary statistics for the analysis sample across all eight years.

Table 2: Summary Statistics — Formal Dependent Workers, ENAHO 2015–2023

| | 2015 | 2016 | 2017 | 2018 | 2019 | 2021 | 2022 | 2023 |
|---|---|---|---|---|---|---|---|---|
| MW (S/.) | 750 | 850 | 850 | 930 | 930 | 930 | 1,025 | 1,025 |
| $N$ | 10,059 | 11,488 | 10,895 | 11,723 | 10,922 | 8,946 | 9,554 | 9,838 |
| Median wage | 1,400 | 1,500 | 1,500 | 1,673 | 1,700 | 1,800 | 1,800 | 1,863 |
| % at MW | 4.6 | 4.8 | 6.0 | 4.8 | 6.5 | 7.8 | 5.7 | 8.2 |
| % below MW | 10.9 | 14.0 | 10.4 | 14.6 | 12.5 | 11.5 | 16.9 | 12.6 |
| Median Kaitz | 0.536 | 0.567 | 0.567 | 0.556 | 0.547 | 0.517 | 0.569 | 0.550 |
| Mean hrs/wk | 46.4 | 46.2 | 46.4 | 46.3 | 46.2 | 46.8 | 46.6 | 46.5 |
| Formality (%) | 26.7 | 27.9 | 27.4 | 27.6 | 27.3 | 23.2 | 24.3 | 26.1 |

*Note*: $N$ = unweighted formal dependent workers with monthly wage S/1–6,000 and positive survey weight. % at MW = share earning MW±25 (one bin). % below MW = share earning below MW. Median Kaitz = MW / median wage, national. Formality rate = formal dependent / all employed. COVID-19 year 2021 shows depressed formality.

Median wages rise from S/1,400 (2015) to S/1,863 (2023), a 33% nominal increase that roughly tracks the 37% MW increase. The Kaitz index ranges from 0.517 to 0.569 with no clear trend; it does not monotonically increase over time. The share below MW rises sharply in pre-event years (14.0% in 2016, 14.6% in 2018, 16.9% in 2022) and then falls as wages adjust to the new floor. Weekly hours are stable at 46–47 hours throughout the period.

# 4 Empirical Strategy

## 4.1 Pre-Post Distributional Estimator

I use a pre-post distributional estimator inspired by the framework of Cengiz et al. (2019), adapted for a national MW setting without control jurisdictions. Cengiz et al. (2019) con-

struct the counterfactual using U.S. states that did not raise their MW. Peru has a single national MW; I use the pre-period wage distribution with a background correction from the upper tail ($> 2 \times \mathrm{MW}_{\mathrm{new}}$) instead. The closer methodological precedent is Harasztosi and Lindner (2019), who study Hungary's national MW.

The wage distribution is discretized into S/25 bins over the range S/0–S/6,000, weighted by survey expansion factors (`fac500a`). For each bin $b$, I compute the share of the workforce in the pre-period ($s_b^{\mathrm{pre}}$) and post-period ($s_b^{\mathrm{post}}$), and compute the raw delta: $\Delta_b = s_b^{\mathrm{post}} - s_b^{\mathrm{pre}}$.

The counterfactual correction uses an inverse-absolute-weighted average of $\Delta_b$ in the "clean zone" (bins with centers above $2 \times \mathrm{MW}_{\mathrm{new}}$), where the MW is unlikely to affect the distribution:

$$\bar{\Delta}_{\mathrm{clean}} = \frac{\sum_{b \in \mathcal{C}} w_b \cdot \Delta_b}{\sum_{b \in \mathcal{C}} w_b}, \quad w_b = \frac{1}{|\Delta_b| + \epsilon}$$

where $\epsilon = 10^{-8}$. The adjusted delta is $\Delta_b^{\mathrm{adj}} = \Delta_b - \bar{\Delta}_{\mathrm{clean}}$.

**Missing mass** (negative-only bins in the affected zone):

$$B^{\mathrm{miss}} = \sum_{\substack{b \in [0.85 \times \mathrm{MW}_{\mathrm{old}},\, \mathrm{MW}_{\mathrm{new}}) \\ \Delta_b^{\mathrm{adj}} < 0}} \left| \Delta_b^{\mathrm{adj}} \right| \tag{1}$$

**Excess mass** (positive-only bins in the excess zone):

$$B^{\mathrm{exc}} = \sum_{\substack{b \in [\mathrm{MW}_{\mathrm{new}},\, \mathrm{MW}_{\mathrm{new}} + 250) \\ \Delta_b^{\mathrm{adj}} > 0}} \Delta_b^{\mathrm{adj}} \tag{2}$$

The **bunching ratio** is $R = B^{\mathrm{exc}} / B^{\mathrm{miss}}$. A ratio of 1 indicates perfect redistribution: all displaced workers reappear above the new MW. A ratio below 1 indicates that some fraction of the missing mass does not reappear, consistent with job loss, informalization, or transitions beyond the S/250 excess window.

This estimator does not require parallel trends and is valid for all three events.

## 4.2 Bootstrap Inference

Standard errors are computed by block-bootstrapping individuals within each pre/post cross-section (1,000 replications). For each bootstrap draw, I resample with replacement and recompute the full bunching ratio. The 95% confidence interval is the 2.5th and 97.5th percentiles of the bootstrap distribution.

## 4.3 Hours DiD

The intensive-margin hours response is measured by a 2×2 DiD of weighted mean weekly hours (`p513t`). The treatment group is workers in the affected wage zone $[0.85\times\mathrm{MW}_{\mathrm{old}}, \mathrm{MW}_{\mathrm{new}})$ in the pre-period and $[\mathrm{MW}_{\mathrm{new}}, 1.3 \times \mathrm{MW}_{\mathrm{new}}]$ in the post-period (tracking where displaced workers land). The control group is workers in $[1.5 \times \mathrm{MW}_{\mathrm{new}}, 3.0 \times \mathrm{MW}_{\mathrm{new}}]$ in both periods.

## 4.4 Failed Identification Strategies (Appendices B–C)

I attempted three regression-based identification strategies for employment effects, all of which fail. The event-study DiD uses department-level Kaitz variation; pre-trends are violated for Events B ($p = 0.007$) and C ($p = 0.017$). The Kaitz IV produces $F$-statistics of 1.5, 2.6, and 0.1—all far below the Stock-Yogo threshold. See Appendices B and C for details.

# 5 Bunching Results

## 5.1 Main Estimates

Table 3 presents the bunching estimates for all three events and three populations.

Table 3: Pre-Post Bunching Estimates — ENAHO Cross-Section

| Event | Group | Missing (pp) | Excess (pp) | Ratio |
|---|---|---|---|---|
| **A** (750→850) | Formal dep | 6.78 | 4.72 | 0.696 |
| | Informal dep | 4.45 | 5.42 | 1.218 |
| | All workers | 5.41 | 4.72 | 0.872 |
| **B** (850→930) | Formal dep | 8.03 | 6.66 | 0.829 |
| | Informal dep | 5.28 | 5.21 | 0.987 |
| | All workers | 6.44 | 5.81 | 0.901 |
| **C** (930→1,025) | Formal dep | 13.02 | 10.80 | 0.830 |
| | Informal dep | 5.50 | 4.46 | 0.811 |
| | All workers | 8.29 | 6.84 | 0.824 |

*Note*: Missing (pp) = share lost from $[0.85 \times \mathrm{MW}_{\mathrm{old}}, \mathrm{MW}_{\mathrm{new}})$ using negative-only bins. Excess (pp) = share gained at $[\mathrm{MW}_{\mathrm{new}}, \mathrm{MW}_{\mathrm{new}} + 250)$ using positive-only bins. Background correction: inverse-abs-weighted mean from clean zone ($> 2 \times \mathrm{MW}_{\mathrm{new}}$). Bins: S/25, range S/0–6,000.

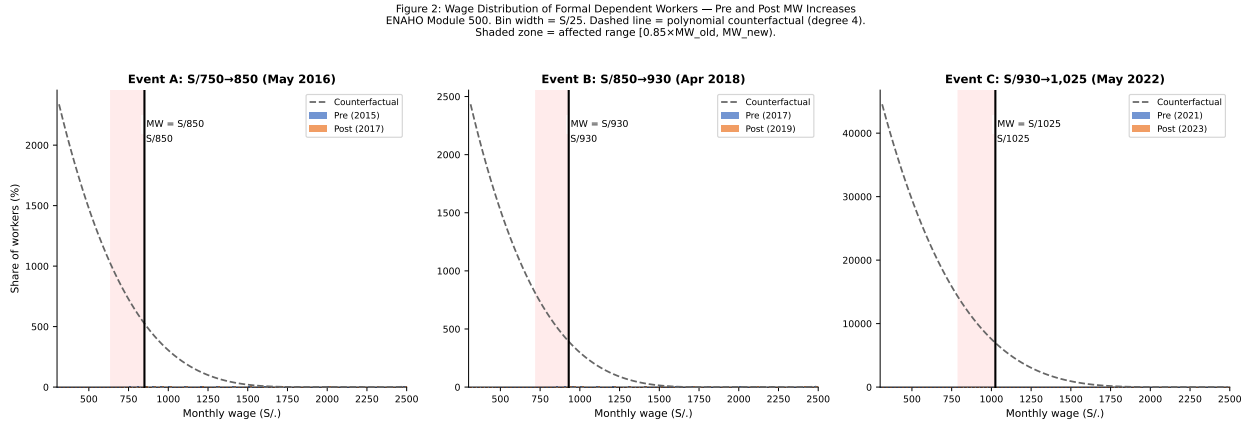


Figure 1: Wage Distributions Before and After MW Increases

*Note*: Three panels show the formal dependent wage distribution (S/25 bins, weighted by `fac500a`) before (blue) and after (orange) each MW increase. Vertical lines mark the old (dashed gray) and new (solid red) MW. Shaded region = affected zone $[0.85 \times \text{MW}_{\text{old}}, \text{MW}_{\text{new}})$.

Formal dependent workers show ratios consistently below 1 across all three events (0.696–0.830), indicating that more workers leave the affected zone than reappear at the new MW. Missing mass increases from 6.78pp (Event A) to 13.02pp (Event C), reflecting the growing MW bite as the RMV rises relative to the wage distribution.

Bootstrap 95% confidence intervals are $[0.567, 0.896]$ (A), $[0.716, 1.016]$ (B), and $[0.716, 0.960]$ (C). Events A and C reject $R = 1$ at the 95% level; Event B does not—approximately 4.7% of bootstrap draws produce $R \geq 1$. For Event B, the data are consistent with both full redistribution and displacement of up to 28% of affected workers.

Informal workers show a ratio above 1 in Event A (1.218), approximately 1 in Event B (0.987), and below 1 in Event C (0.811). There is no consistent spillover pattern.

## 5.2 Placebo Test

Applying the same estimator to artificial thresholds on the Event B population—fake increases from S/1,100→S/1,200 and S/1,400→S/1,500—produces ratios of 0.114 and 0.013, with missing mass 7 times smaller than at the true MW (1.2pp and 2.0pp vs. 8.0pp). The

bunching signal is specific to the actual MW threshold.

### 5.3 Wage Compression

Table 4 presents the wage compression DiD with a decomposition into mechanical (composition) and genuine components.

Table 4: Wage Compression DiD with Composition Decomposition

| Event | Total DiD | Mechanical | Genuine | % Mech. | New arrivals |
|---|---|---|---|---|---|
| A (750→850) | −0.035 | −0.028 | −0.008 | 78% | 13% of zone |
| B (850→930) | −0.050 | −0.021 | −0.029 | 41% | 10% of zone |
| C (930→1,025) | −0.069 | −0.064 | −0.005 | 92% | 26% of zone |

*Note*: Compression zone = $[\mathrm{MW}_{\mathrm{new}}, 1.5 \times \mathrm{MW}_{\mathrm{new}}]$; high-wage zone = $[2 \times \mathrm{MW}_{\mathrm{new}}, 3 \times \mathrm{MW}_{\mathrm{new}}]$. Mechanical = predicted DiD from new arrivals entering at $\log(\mathrm{MW}_{\mathrm{new}})$ while existing workers grow at the high-wage rate. Genuine = total minus mechanical. With individual controls (age, sex, education, sector), the compression coefficient is −0.045 ($p < 0.001$).

For Event B, approximately 41% of the −0.050 DiD reflects the composition effect—workers pushed from below the MW to the new floor entering the compression zone and diluting its mean log wage. The remaining 59% (−0.029 log points) represents genuine relative wage compression. For Events A and C, the composition effect dominates (78% and 92%). The compression DiD is a descriptive statistic of distributional change, not a causal estimate.

## 6 Heterogeneity

Table 5 through Table 7 present bunching ratios across subgroups. In all panels, the Total row (0.696 / 0.829 / 0.830) matches Table 3 exactly.

Table 5: Bunching Ratios by Industry Sector

| Sector | Event A | Event B | Event C | Average |
|---|---|---|---|---|
| **Total** ($N = 9{,}952$) | **0.696** | **0.829** | **0.830** | **0.785** |
| Agriculture/Mining/Utilities ($N = 1{,}031$) | 1.641 | 0.765 | 0.407 | 0.938 |
| Manufacturing ($N = 904$) | 1.000 | 0.869 | 1.049 | 0.973 |
| Commerce ($N = 937$) | 0.784 | 1.209 | 0.723 | 0.905 |
| Transport/hospitality ($N = 556$) | 1.041 | 1.018 | 0.585 | 0.881 |
| Finance/professional services ($N = 1{,}113$) | 0.768 | 1.018 | 0.806 | 0.864 |
| Public administration ($N = 1{,}639$) | 1.033 | 0.498 | 0.804 | 0.778 |
| Education and health ($N = 3{,}021$) | 0.598 | 0.709 | 0.975 | 0.760 |

*Note*: $N = 2015$ pre-period sample. Construction excluded (sparse cells, $< 500$ obs). Agriculture/Mining includes CIIU rev4 codes 35–39 (utilities). Sector from `p506r4` (CIIU 2-digit).

Table 6: Bunching Ratios by Firm Size, Age, and Sex

| Group | Event A | Event B | Event C | Average |
|---|---|---|---|---|
| *Panel B: Firm size* | | | | |
| Micro ($\leq 10$, $N = 705$) | 0.899 | 0.909 | 0.690 | 0.833 |
| Small (11–50, $N = 1{,}238$) | 0.919 | 0.901 | 0.598 | 0.806 |
| Medium/Large (51+, $N = 7{,}557$) | 0.686 | 0.793 | 0.896 | 0.792 |
| *Panel C: Age and sex* | | | | |
| Age 18–24 | 0.661 | 1.010 | 0.805 | 0.826 |
| Age 25–44 | 0.748 | 0.792 | 0.797 | 0.779 |
| Age 45–64 | 0.701 | 0.927 | 0.853 | 0.827 |
| Men | 0.726 | 0.788 | 0.778 | 0.764 |
| Women | 0.697 | 0.922 | 0.877 | 0.832 |

*Note*: Firm size from `p512b` (establishment worker count). Workers with missing `p512b` (4.3–5.8% per year) are excluded from Panel B; the three rows sum to 95.5% of the sample.

Table 7: Bunching Ratios by Geography and Employer Sector

| Group | Event A | Event B | Event C | Average |
|---|---|---|---|---|
| Lima Metropolitan ($N = 2{,}594$) | 0.618 | 1.050 | 0.817 | 0.828 |
| Rest of country ($N = 7{,}358$) | 0.856 | 0.621 | 0.888 | 0.788 |
| Urban ($N = 9{,}286$) | 0.702 | 0.843 | 0.828 | 0.791 |
| Rural ($N = 666$) | 1.016 | 0.550 | 0.909 | 0.825 |
| Public sector ($N = 4{,}315$) | 0.921 | 0.456 | 0.865 | 0.747 |
| Private sector ($N = 5{,}576$) | 0.766 | 0.962 | 0.768 | 0.832 |

*Note*: Public sector identified by `p510` $\in \{1, 2, 3\}$ or CIIU $\in \{84, 85, 86\}$.

Ratios cluster near 0.80 across most subgroups. The private sector absorbs MW in-

creases more effectively (average ratio 0.832) than the public sector (0.747), where rigid pay scales limit adjustment. Manufacturing consistently shows the highest ratios ($\approx 0.97$), likely reflecting higher enforcement density. No subgroup shows dramatically different behavior.

**Ethnicity.** Among formal dependent workers, indigenous (Quechua/Aymara/native-language) and non-indigenous workers have similar wage distributions (both median S/1,400) and similar MW exposure (8.5% vs. 10.7% below MW). The ethnic divide operates through access to formal employment: the indigenous formality rate is 5.7% versus 20.7% for Spanish speakers (a 3.6× gap). The 925 indigenous formal dependent workers are predominantly public-sector employees (59%), concentrated in highland departments. Bunching ratios by ethnicity are unstable across events and reflect sector composition rather than differential MW compliance.

# 7 Hours

Table 8 presents the intensive-margin hours DiD.

Table 8: Weekly Hours DiD

| Event | Treat pre | Treat post | Δ treat | Ctrl pre | Ctrl post | Δ ctrl | DiD |
|---|---|---|---|---|---|---|---|
| A (750→850) | 44.4h | 44.8h | +0.4h | 42.1h | 41.5h | −0.7h | +1.1h |
| B (850→930) | 44.8h | 45.1h | +0.3h | 41.2h | 42.0h | +0.8h | −0.5h |
| C (930→1,025) | 44.5h | 46.4h | +1.9h | 42.1h | 43.6h | +1.5h | +0.4h |

*Note*: Treatment pre = workers in $[0.85 \times \mathrm{MW_{old}}, \mathrm{MW_{new}})$; treatment post = workers in $[\mathrm{MW_{new}}, 1.3 \times \mathrm{MW_{new}}]$ (bunched-up zone). Control = $[1.5 \times \mathrm{MW_{new}}, 3.0 \times \mathrm{MW_{new}}]$. Weighted means using `fac500a`. Variable: `p513t` (total weekly hours).

No DiD exceeds ±1.1h/week. There is no evidence of intensive-margin hours adjustment in response to MW increases.

# 8 Conclusion

**Employment effects cannot be identified.** The event study fails pre-trends for Events B ($p = 0.007$) and C ($p = 0.017$); Event A has no testable pre-period. The IV approach fails due to weak first stages ($F < 3$ in all events). The bunching ratios (0.70–0.83, bootstrap CIs [0.57–1.02]) are consistent with both modest displacement (up to 28% of affected workers) and full redistribution. The data cannot distinguish between these scenarios.

**What can be concluded**: Peru's MW increases produce a clear, placebo-verified redistribution of the formal wage distribution. Workers near the MW floor receive raises—the pre-post estimator documents 6.78–13.02 percentage points of missing mass below the old floor across three events, with 70–83% reappearing as excess mass at the new floor. The intensive margin (hours) shows no adjustment. Compression of 3–7 percentage points is present but 41–92% is mechanical composition. The residual genuine compression ($-0.005$ to $-0.029$ log points) represents modest spillover above the MW threshold; with individual controls, the compression coefficient is $-0.045$ ($p < 0.001$).

Three points of external validity strengthen these findings. First, EPE Lima quarterly data produces ratios (0.73–1.03) consistent with the ENAHO results despite different formality definitions, event windows, and geographic scope. Second, placebo tests at artificial thresholds produce ratios 7 times smaller. Third, informal workers show partial response in Event A (1.218) but not in Events B–C, suggesting some spillover at lower MW levels.

This is not a data-quality failure—it is a structural feature of single-national-MW settings. Future work using administrative wage registries (SUNAT, ESSALUD) that allow quarterly firm-level identification could substantially improve power.

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# A ENAHO Panel 978 — Event C

The ENAHO panel 978 tracks a subsample of individuals from 2021 to 2023. Treatment: formal dependent workers in 2021 with wages in [S/791, S/1,025). Control: wages in [S/1,230, S/2,563]. Approximately 76% of both groups are not re-interviewed; results apply to the surviving 24% only. Outcome statistics are weighted by `facpanel2123`. Attrited workers are *not* counted as non-employed.

Table 9: Panel Transition Results (Event C, 2021→2023)

| Metric | Treatment ($N = 1{,}171$) | Control ($N = 3{,}903$) | DiD |
|---|---|---|---|
| Re-interviewed rate | 24.4% | 23.6% | — |
| *Among re-interviewed (weighted):* | | | |
| Employment retention | 63.9% | 60.1% | +3.8pp |
| Formal dep retention | 46.8% | 49.9% | −3.1pp |
| Δ log wage (stayers) | +0.307 | +0.154 | +0.153 |
| Δ hours (stayers) | +0.7h | −1.3h | +2.0h |

*Caveat*: 76% attrition with differential selection ($p = 0.022$). Do not interpret causally.

# B Event Study — Pre-Trend Violations

All three events have identification problems. Event A: no prior years available (2015 = base); parallel trends untestable. Event B: joint pre-trend $F$-test $p = 0.007$; higher-Kaitz departments had systematically higher employment before the MW change. Event C: joint pre-trend $p = 0.017$; pre-COVID Kaitz–employment correlation renders the comparison non-parallel.

# C IV / OWE — Not Identified

Three IV approaches were tested. (1) Department-level Kaitz IV: $F = 1.5/2.6/0.1$ across events, all far below the Stock-Yogo threshold. (2) Province-level IV: $F = 11.72$ for Event B but with reversed first-stage sign and failed pre-trends. (3) Within-year province IV: $F = 16.15$ but the placebo (no-MW-change year) produces $F = 12.71$—the instrument captures seasonal wage dynamics, not MW exposure.

Table 10: Department-Level Kaitz IV

| Event | $\hat{\pi}$ (FS) | SE | $F$ | $\hat{\beta}$ (RF) | SE | OWE | SE |
|---|---|---|---|---|---|---|---|
| A | +0.169 | 0.082 | 1.5 | +0.014 | 0.038 | +0.082 | 0.229 |
| B | −0.070 | 0.103 | 2.6 | +0.122 | 0.044 | — | — |
| C | +0.132 | 0.090 | 0.1 | +0.220 | 0.076 | +1.667 | 1.271 |

*Note*: Event B OWE not reported (reversed first-stage sign). SE clustered by department (24 clusters). All $F < 3$; instrument is weak.

# D EPEN CIU Annual 2023 — Lee-Saez Bunching

Single-period Lee-Saez estimate from INEI code 873 ($N = 53{,}316$ formal urban workers). All cities show excess mass at MW: ratio 1.622. These are not directly comparable to the pre-post ratios in Table 3.

# E EPE Lima Quarterly Bunching

EPE Lima provides an independent dataset check with tighter 6-month event windows (Q1 vs. Q3 of each event year). Formality uses `p222 == 1` (EsSalud only), which is narrower than ENAHO. Lima Metropolitan Area only.

Table 11: EPE Lima Pre-Post Bunching (6-Month Windows)

| Event | Window | $N$ pre | $N$ post | Near-MW | Ratio | 95% CI | ENAHO |
|---|---|---|---|---|---|---|---|
| A | Q1→Q3 2016 | 2,622 | 2,599 | 240 | 1.031 | [0.70, 1.63] | 0.696 |
| B | Q1→Q3 2018 | 2,280 | 2,438 | 302 | 0.733 | [0.60, 1.03] | 0.829 |
| C | Q1→Q3 2022 | 1,933 | 1,936 | 331 | 0.885 | [0.70, 1.13] | 0.830 |

*Note*: All ENAHO ratios fall within EPE Lima bootstrap CIs. Event A diverges in point estimates; Lima's lower Kaitz may explain the higher absorption rate. CIs wide due to small $N$.